Lesson 9:
Improving Your Agent Cost

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Last time / today

Analyze

- L1: built the agent from a spec
- L2: traced it in Langfuse
- L3: generated synthetic scenarios
- L4: found failure modes in traces

Measure

- L5: LLM judges checked against human labels
- L6: retrieval metrics and eval cases in CI
- L7: monitoring and agent safety

Improve

- L8: obvious fixes, autoresearch, GEPA
- **L9 (today): cost**

Lecture outline

1. **The cost and accuracy frontier:** where the money goes, and which changes to keep
2. **Do less work:** fewer tokens, fewer agent steps
3. **Make the remaining work cheaper:** caching and cascades
4. **Changing the weights, and updating your models**

Whiteboard: the cost and accuracy frontier

Strategy 1: Do Less Work

Strategies

- Shorter outputs
- Smaller tool results
- Less retrieval depth
- Workflow instead of agent, where it passes the suite
- Batch async work: judges, backfills, reports

Sources

Anthropic, 2024, Building Effective Agents
Workflows vs. agents; start with the simplest design

Chroma, 2025, Context Rot
All 18 models tested got worse as input grew

Nayab et al., 2024, Concise Thoughts
Limiting output length cut cost, sometimes raised accuracy

OpenAI Batch API
50% discount, results within 24 hours

Strategy 2: Make the Remaining Work Cheaper

Strategies

- **Prompt caching:** static content first, fixed tool order, append only
- **Model cascade:** cheap model first, escalate when uncertain
- **Routing:** send each request to one model up front

Sources

Lumer et al., 2026, Don't Break the Cache
Caching on long agentic tasks cut API cost 45 to 80%

Manus, 2025, Context Engineering for AI Agents
Cache hit rate as the key production metric

Jung et al., 2025, Trust or Escalate
Cascade of LLM judges with guaranteed human agreement, up to 87% cheaper

Shreya Shankar, How to Cut Your LLM Classification Costs by 90% (video)
Building a model cascade step by step

Chen et al., 2023, FrugalGPT
Cascade across models, up to 98% cheaper

Ong et al., 2024, RouteLLM
Router trained on preference data

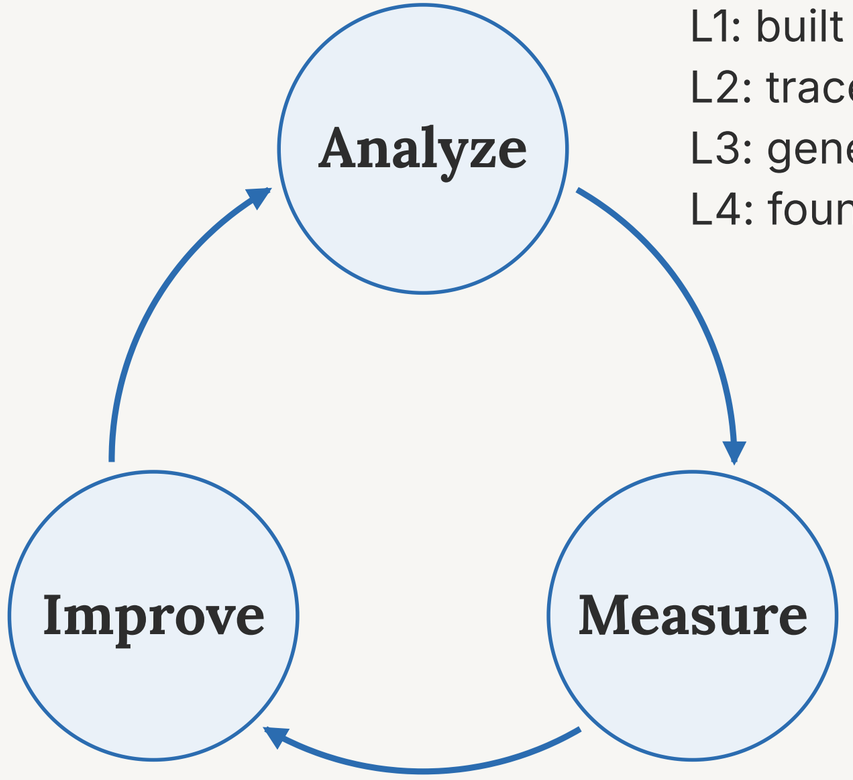
When to change the weights

- Only after prompt changes, cutting work, caching, and cascades stop helping
- **Distillation / fine tuning:** train a small model on the strong model's passing traces
  - Hsieh et al., 2023, Distilling Step-by-Step
- **RL:** only with frequent, verifiable rewards
  - Lambert et al., 2024, Tülu 3 (RLVR)
- Run the trained model on the test cases and compare its cost and score with the other configurations
- **Don't do it unless you really need to!**

Updating your models

- Swap the new model into each saved configuration and rerun the suite
- **Prompt drift:** the same prompt behaves differently on a new model version
  - Chen, Zaharia, and Zou, 2023: GPT-4 accuracy on identifying prime numbers fell from about 98% to about 2% in three months
- Hopefully less of a problem as models get better
- Newer models do more of the right behavior without being told
- **When you switch models, remove prompt instructions the new model no longer needs**

Recap



L1: built the agent from a spec
L2: traced it in Langfuse
L3: generated synthetic scenarios
L4: found failure modes in traces

L5: judges vs. human labels
L6: eval cases in CI
L7: monitoring and safety

L8: autoresearch, GEPA
L9: cost

Homework 9 and thank you!

Homework 9

- **Part A:** cost per call site, agent and judges
- **Part B:** one change that cuts tokens
- **Part C:** prompt caching, before and after
- **Part D:** model cascade on one judge
- **Part E:** frontier of your configurations

You are now an AI evals expert! 🎉

- Thank you to our amazing TAs, Antaripa and Eleanor!
  - Antaripa: x.com/doesdatmaksense
  - Eleanor: x.com/intellectronica
- Office hours today, 12:15 PM PT
- Discord stays open after the course
- Homework 9 instructions
- Course reader